

Qatar-2b

R-1191

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_160404 · created 2026-07-31T05:43:59+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457482.85947 +/- 0.00026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.066
Transit depth (Rp/Rs)^2	3.8e-06 +/- 0.0026 [%]
Orbital Inclination (inc)	87.98 +/- 1.73
Airmass coefficient 1 (a1)	0.49 +/- 0.32
Airmass coefficient 2 (a2)	0.43 +/- 0.43
Scatter in the residuals of the lightcurve fit is	61.13 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	7.5
Transit Duration (day)	0.0666 +/- 0.0043

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.0 $\pm$ 0.066 vs 0.16327 (deviation 2.32 σ)
Duration fit vs published	0.0666 d vs 0.0767 d (deviation 2.2 σ)
Mid-time O-C	0.5 min
Depth SNR	0.0
Residual scatter	61.13 %

Light curve

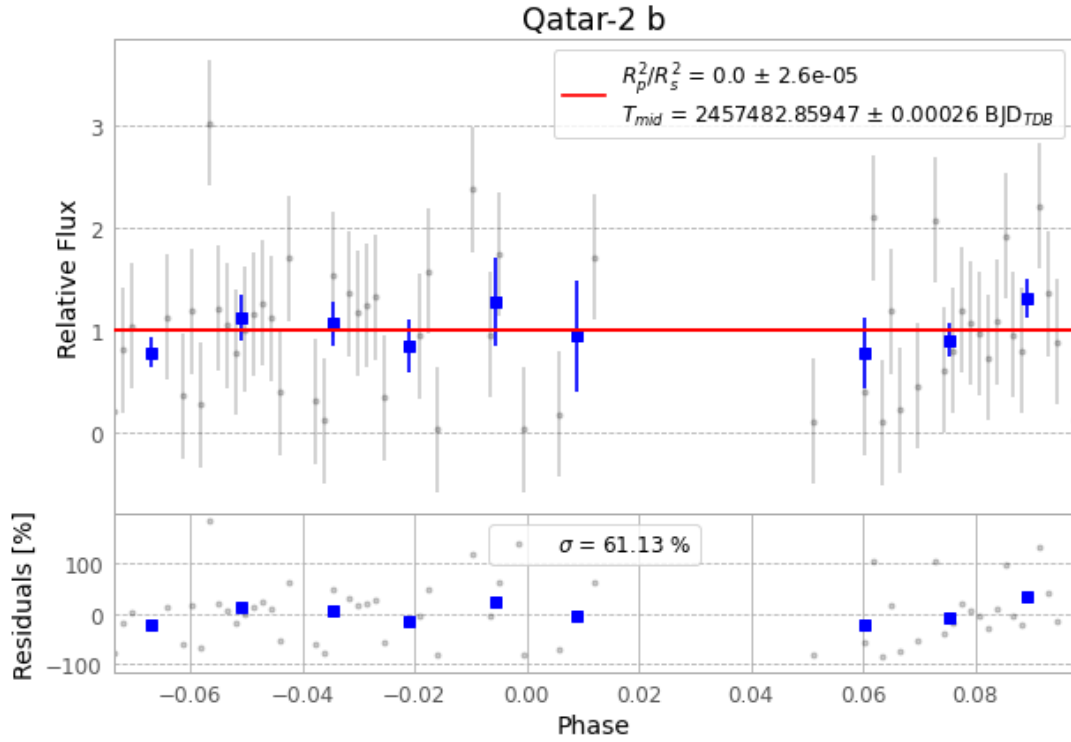


Figure 1 — FinalLightCurve_Qatar-2 b_2016-04-03.png (SHA-256 2961e6db57115408...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [329, 479], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f6d924cbc0f1eaf3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

116 FITS frames (76 MB), Qatar-2160404055112.FITS → Qatar-2160404113612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-160404.log (ce3bbd651240...), FinalLightCurve_Qatar-2 b_2016-04-03.png (2961e6db5711...), FinalLightCurve_Qatar-2 b_2016-04-03.csv (fb26c4c27028...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 05:44Z.